

OrthoHMM: HMM-Centered Group Inference With Phylogenetic Refinement

Condensed scientific draft, 27 September 2026. Not submission-ready. The [extended manuscript](#) retains detailed methods, historical analyses, citations and audit records. This main text does not supersede frozen protocols or historical result manifests.

Abstract

Orthology inference requires balancing homolog recovery against separation of paralogs. We evaluated an HMM-centered pipeline with candidate-family expansion and phylogenetic refinement against established orthology tools. QfO and OrthoBench were the primary development-exposed benchmarks; a frozen YGOB evaluation assessed transfer to additional taxa. Phylogenetic OrthoHMM improved on its high-sensitivity configuration but did not consistently outperform full OrthoFinder. On OrthoBench, their group-recovery F1 values were 74.11% and 72.74%, respectively. On corrected QfO, OrthoFinder had higher VGNC, SwissTrees and TreeFam-A point estimates, whereas OrthoHMM had higher GO/EC similarity and FAS. A matched-recall simulation control supported an initial HMM-search contribution in a fixed graph procedure, not full-pipeline superiority. Independent-family generalization, uncertainty for several QfO endpoints and controlled comparative timing remain unresolved. The evidence supports a bounded contribution and an explicit precision-recall trade-off, not universal accuracy or efficiency claims.

Methods

The retained configurations are high-sensitivity OrthoHMM and the `satellite_v2` phylogenetic pipeline. The original OrthoHMM preprint (Steenwyk et al. 2024) describes its lineage, not all subsequent implementation changes. The prospective method was frozen at `7f3a9e4`, with BLOSUM62, E-value threshold $1e-4$, Leiden CPM resolution 0.1 and seed 4 (Traag, Waltman, and van Eck 2019; Traag, Van Dooren, and Nesterov 2011). The phylogenetic configuration expands candidate families, infers gene and species trees, and applies positive-paralogy pair inference. Historical runs retain their actual revisions rather than inheriting this prospective pin. The [method diagram](#) distinguishes initial search, profile refinement, candidate expansion and phylogenetic inference.

Comparators were OrthoFinder 3.1.5 (David M. Emms et al. 2026a, 2026b), SonicParanoid 2.0.9 (Cosentino, Sriswasdi, and Iwasaki 2024), ProteinOrtho 6.3.6 (Klemm, Stadler, and Lechner 2023), FastOMA 0.3.5 (Majidian et al. 2025) and OrthoMCL 1.4 (Li, Stoeckert, and Roos 2003). These method references do not replace run-specific executable provenance. OrthoFinder's sequence-only MCL checkpoint was a diagnostic output, not a separately finalized phylogenetic analysis. FastOMA used a supplied OrthoFinder species tree. QfO inputs included native ortholog pairs, native post-clustering relations or group-derived cross-species pairs as appropriate; these are not interchangeable output semantics. The [generated comparison](#) reports each conversion and prediction count.

OrthoBench measures curated group recovery (David M. Emms and Kelly 2020). QfO (Quest for Orthologs consortium et al. 2016; Altenhoff et al. 2020) reports GO and EC similarity, VGNC, SwissTrees and TreeFam-A F1, and FAS separately. Their arithmetic mean is a project-defined secondary summary, not an official QfO F1. Three Kingdoms is supplementary BUSCO-reference recovery (Manni et al. 2021), not genome-wide orthology truth. QfO and OrthoBench were repeatedly inspected during development. The [YGOB protocol](#), using the curated homology/syntenic resource (Byrne and Wolfe 2005), froze evaluation before test-score inspection; overlap assessment limits its interpretation to novel-taxon transfer rather than family-disjoint validation.

Uncertainty analyses resample declared reference units and recompute the benchmark statistic within replicates. They retain prespecified multiplicity adjustments, negative findings and unavailable contrasts. Curated families are not automatically independent: shared evolutionary history and predictions joining families can violate exchangeability. Dependent protein pairs are not treated as independent observations to obtain narrower intervals.

Results

Accuracy Depends On The Endpoint

Phylogenetic OrthoHMM achieved 74.1061% OrthoBench F1 versus 72.7365% for full OrthoFinder. A fresh installed full run reproduced all 59,770 OrthoHMM groups exactly; this establishes reproducibility, not independent accuracy or a controlled runtime comparison. [Full-run verification](#).

On corrected QfO, phylogenetic OrthoHMM versus full OrthoFinder scored 0.901690 versus 0.988546 on VGNC, 0.833513 versus 0.848413 on SwissTrees, and 0.614864 versus 0.791918 on TreeFam-A. OrthoHMM had higher GO similarity (0.490349 versus 0.469548), EC similarity (0.965650 versus 0.936130) and FAS (0.762993 versus 0.691422). These endpoints differ in reference scope and eligibility; they do not form a single accuracy ranking. [All eight methods](#).

For corrected SwissTrees, phylogenetic-minus-full-OrthoFinder F1 was -1.4900 percentage points, with a multiplicity-adjusted interval of [-8.7078, 7.2090]. Phylogenetic-minus-high-sensitivity OrthoHMM was +14.8015 points [4.9791, 26.6015]. These conditional estimates use 18 development-exposed families and a 24-endpoint adjustment; inclusion of zero is not equivalence. Candidate expansion also changes between OrthoHMM configurations, so their contrast does not isolate reconciliation. [Protocol and contrast evidence](#).

All 56 method-pair GO/EC comparisons had identical six-decimal scores on shared scored pairs. Aggregate differences arose from different eligible pair sets and their denominators. Restricting evaluation to intersections would change the endpoint rather than resolve uncertainty. Historical raw hashes were verified for all 16 input tables. [Scored-pair audit](#).

Component Evidence Is Bounded

The corrected QfO factorial supported candidate-expansion-by-reconciliation interactions on SwissTrees F1, while all four profile-refinement F1 intervals included zero. Reconciliation increased precision and reduced recall. These results support conditional component effects, not proof that profile expansion universally improves orthology inference. [Complete factorial](#).

In 35 matched-recall simulation datasets spanning seven conditions and five seeds, HMM-derived search evidence yielded mean downstream graph F1 of 83.6097%, versus 80.5180% for DIAMOND (Buchfink, Reuter, and Drost 2021). The adjusted paired seed-block interval for the +3.0917-point difference was [1.6782, 4.5313]. Profile expansion and phylogeny were off. This is a development-exposed fixed-graph comparison, not a comparison against OrthoFinder or a matched-effort result. Score rankings and hit identities differ, so the design does not isolate a causal mechanism. Matched-recall control.

Transfer And Biological Recovery Reveal Trade-Offs

Frozen YGOB group-recovery F1 was 92.233654% for phylogenetic OrthoHMM and 92.318524% for full OrthoFinder. Their difference was -0.084870 points, with adjusted interval [-0.622528, 0.445222]. OrthoHMM had higher precision and lower recall. The evaluation projects predictions onto the reference universe and assumes exchangeable pillars; it does not establish unrestricted generalization or resolved pairwise orthology. YGOB results.

The prespecified whole-genome-duplicate application used experimental evidence from Kuzmin and colleagues (Kuzmin et al. 2020) and assessed paralog separation together with retention of reference homologs. Phylogenetic OrthoHMM's greater separation than high sensitivity did not remove its homolog-recovery losses. Full OrthoFinder remained an important negative comparison. Raw separation alone is insufficient because splitting off anchors can increase separation without useful ortholog recovery. The complete application, controls and subsequent traces remain in the extended manuscript rather than selecting only favorable examples.

Discussion And Limitations

The supported contribution is an HMM-centered alternative with measurable component effects and explicit accuracy trade-offs. Initial HMM search has bounded support in matched-recall simulations; additional profile refinement has not demonstrated a general benefit. Neither simulation evidence nor an OrthoBench point advantage establishes superiority over full OrthoFinder.

Original TreeFam-A family mappings remain unavailable. VGNC dependence and rare-error diagnostics do not justify the candidate confidence-interval procedure. Valid paired uncertainty for GO/EC, FAS and the secondary mean is also unfinished. These gaps must remain visible alongside point estimates. The corrected FAS sample audit reproduces all eight means, but scored fractions range from 0.0067% to 58.46% of reported eligible pairs and every sample reuses proteins across pairs. Native pair-level standard errors do not resolve this comparison uncertainty. The completed simulations and tree perturbations do not cover arbitrary evolutionary conditions, and novel-taxon YGOB testing retains homolog-family overlap with development data.

Recovered OrthoMCL results retain sequence-specific BLAST failures. Failed high-CPM experiments remain unavailable, not zero-scoring or successful runs. Shared-host runs and the historical DGX panel are descriptive resource records; the replacement dedicated timing panel has not run. No controlled speedup is claimed. Versioned archival release, complete dependency/data redistribution review, journal formatting and final visual review remain open.

Reproducibility And Availability

The extended manuscript links frozen protocols, source revisions, seeds, input/output manifests, conversion audits, executable workflows and generated figures. Installed inference and independent readback have been validated on full OrthoBench, but that is not proof of all-method cross-host portability. Historical locks are retained as provenance; patched replacement environments must be distinguished from the binaries used for original scores. The [progress ledger](#) records completed work and unmet requirements. No submission-ready release or archival DOI is claimed.

References

The bibliography attributes the methods and resources cited above. Full execution evidence and additional dependency references remain in the extended manuscript; this selected list is not complete software attribution.

- Altenhoff, Adrian M, Javier Garrayo-Ventas, Salvatore Cosentino, David Emms, Natasha M Glover, Ana Hernández-Plaza, Yannis Nevers, et al. 2020. “The Quest for Orthologs Benchmark Service and Consensus Calls in 2020.” *Nucleic Acids Research* 48 (W1): W538–45. <https://doi.org/10.1093/nar/gkaa308>.
- Buchfink, Benjamin, Klaus Reuter, and Hajk-Georg Drost. 2021. “Sensitive Protein Alignments at Tree-of-Life Scale Using DIAMOND.” *Nature Methods* 18 (4): 366–68. <https://doi.org/10.1038/s41592-021-01101-x>.
- Byrne, Kevin P., and Kenneth H. Wolfe. 2005. “The Yeast Gene Order Browser: Combining Curated Homology and Syntenic Context Reveals Gene Fate in Polyploid Species.” *Genome Research* 15 (10): 1456–61. <https://doi.org/10.1101/gr.3672305>.
- Cosentino, Salvatore, Sira Sriswasdi, and Wataru Iwasaki. 2024. “SonicParanoid2: Fast, Accurate, and Comprehensive Orthology Inference with Machine Learning and Language Models.” *Genome Biology* 25 (1). <https://doi.org/10.1186/s13059-024-03298-4>.
- Emms, David M, and Steven Kelly. 2020. “Benchmarking Orthogroup Inference Accuracy: Revisiting Orthobench.” *Genome Biology and Evolution* 12 (12): 2258–66. <https://doi.org/10.1093/gbe/evaa211>.
- Emms, David M., Yi Liu, Laurence Belcher, Jonathan Holmes, and Steven Kelly. 2026a. “OrthoFinder: Improved Phylogenetic Orthology Inference with Enhanced Accuracy and Scalability.” *Nature Methods* 23 (7): 1327–33. <https://doi.org/10.1038/s41592-026-03126-6>.
- . 2026b. “Author Correction: OrthoFinder: Improved Phylogenetic Orthology Inference with Enhanced Accuracy and Scalability.” *Nature Methods*, August. <https://doi.org/10.1038/s41592-026-03238-z>.
- Klemm, Paul, Peter F. Stadler, and Marcus Lechner. 2023. “Proteinortho6: Pseudo-Reciprocal Best Alignment Heuristic for Graph-Based Detection of (Co-)orthologs.” *Frontiers in Bioinformatics* 3 (December). <https://doi.org/10.3389/fbinf.2023.1322477>.
- Kuzmin, Elena, Benjamin VanderSluis, Alex N. Nguyen Ba, Wen Wang, Elizabeth N. Koch, Matej Usaj, Anton Khmelinskii, et al. 2020. “Exploring Whole-Genome Duplicate Gene Retention with Complex Genetic Interaction Analysis.” *Science* 368 (6498). <https://doi.org/10.1126/science.aaz5667>.
- Li, Li, Christian J. Stoeckert Jr., and David S. Roos. 2003. “OrthoMCL: Identification of Ortholog Groups for Eukaryotic Genomes.” *Genome Research* 13 (9): 2178–89. <https://doi.org/10.1101/gr.1224503>.
- Majidian, Sina, Yannis Nevers, Ali Yazdizadeh Kharrazi, Alex Warwick Vesztrocy, Stefano Pascarelli, David Moi, Natasha Glover, Adrian M. Altenhoff, and Christophe Dessimoz. 2025. “Orthology

- Inference at Scale with FastOMA.” *Nature Methods* 22 (2): 269–72.
<https://doi.org/10.1038/s41592-024-02552-8>.
- Manni, Mosè, Matthew R Berkeley, Mathieu Seppey, Felipe A Simão, and Evgeny M Zdobnov. 2021. “BUSCO Update: Novel and Streamlined Workflows Along with Broader and Deeper Phylogenetic Coverage for Scoring of Eukaryotic, Prokaryotic, and Viral Genomes.” *Molecular Biology and Evolution* 38 (10): 4647–54. <https://doi.org/10.1093/molbev/msab199>.
- Quest for Orthologs consortium, Adrian M Altenhoff, Brigitte Boeckmann, Salvador Capella-Gutierrez, Daniel A Dalquen, Todd DeLuca, Kristoffer Forslund, et al. 2016. “Standardized Benchmarking in the Quest for Orthologs.” *Nature Methods* 13 (5): 425–30.
<https://doi.org/10.1038/nmeth.3830>.
- Steenwyk, Jacob L, Thomas J. Buida III, Antonis Rokas, and Nicole King. 2024. “OrthoHMM: Improved Inference of Ortholog Groups Using Hidden Markov Models.” Preprint. openRxiv.
<https://doi.org/10.1101/2024.12.07.627370>.
- Traag, V. A., P. Van Dooren, and Y. Nesterov. 2011. “Narrow Scope for Resolution-Limit-Free Community Detection.” *Physical Review E* 84 (1). <https://doi.org/10.1103/PhysRevE.84.016114>.
- Traag, V. A., L. Waltman, and N. J. van Eck. 2019. “From Louvain to Leiden: Guaranteeing Well-Connected Communities.” *Scientific Reports* 9 (1). <https://doi.org/10.1038/s41598-019-41695-z>.